

UNIT-1 **Introduction**

(2 hours)

- 1.1 Micro, macro
- 1.2 Engineering economics (See from Book- Damodar Adhikari)
- 1.2 Terminology related to engineering economics (See from Book- Damodar Adhikari)
- 1.3 Economic decision and role of engineers in decision making (See from Book- DA)
- 1.4 Cash flow and cash flow diagram (See from Book- DA)

Micro and Macro Economics

Basis	Microeconomics	Macroeconomics
Meaning	Branch of economics that studies the behavior of an individual consumer, firm, family as well.	Branch of economics that studies the behavior of the whole economy.
Deals	Individual economic issues like demand, supply, pricing of goods/ factor, production, consumption, economic welfare, etc.	Aggregate economic issues like, national income, general price level, employment, trade cycle etc.
Business Application	Applied to operational or internal issues.	Applied environment and external issues as well.
Tools	Demand and Supply.	Aggregate Expenditure and Aggregate output.
Assumption	Long-run, full-employment, perfect competition, passive government role.	Short-run, under full-employment, active role of government in economy.
Importance	Helpful in determining the prices of a product along with the prices of factors of production (land, labor, capital, entrepreneur etc.) within the economy.	Maintains stability in the general price level and resolves the major problems of the economy like inflation, deflation, unemployment and poverty as a whole.
Limitations	It is based on unrealistic assumptions; full employment and perfect competition in the society which are not at all possible.	It is based on average result and fallacy in comparison. It doesn't prove that what is true for aggregate may not be true for individuals.



UNIT-2 Market Economics

(3 hours)

- 2.1 Market, demand, supply and relationship
- 2.2 Elasticity, application of elasticity and government policies
- 2.3 Externality and market inefficiency
- 2.4 Market failure and firm behavior

2.1 Market, demand, supply and relationship

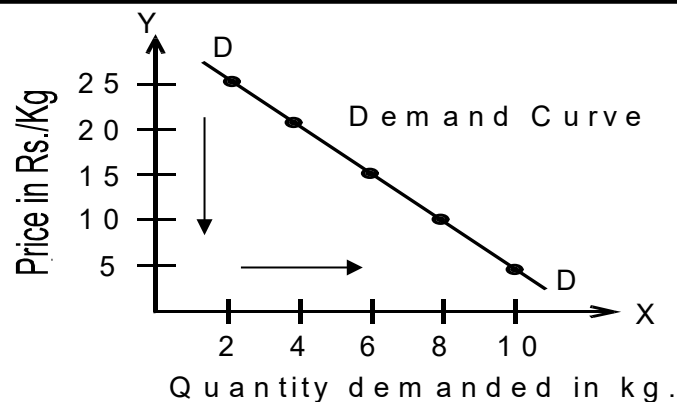
Demand

Demand in economics is not just as a desire to want something. It includes both the willingness and ability of consumers to pay for a goods or service. The demand will be effective when consumer must want the good and also have the financial means to buy it.

Law of Demand

It is defined as *"all other things remaining the same, the quantity demanded of a commodity increase when its price decrease and vice versa."*

Price in Rs./Kg	Quantity demanded in Kg
5	10
10	8
15	6
20	4
25	2



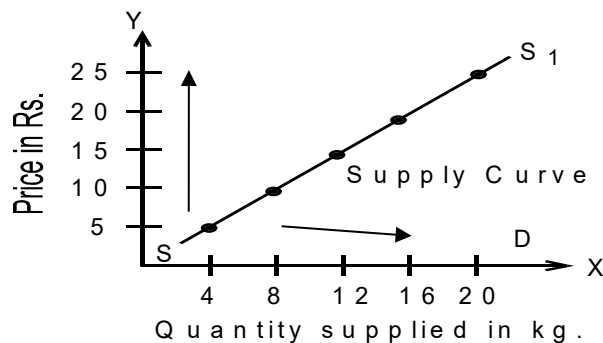
Supply

In economics, supply refers to the quantity of a product available in the market for sale at a specified price and time. It can be defined as the willingness of a seller to sell the specified quantity of a product within a particular price and time period. Here, it should be noted that demand is the willingness of a buyer, while supply is the willingness of a supplier.

Law of Supply

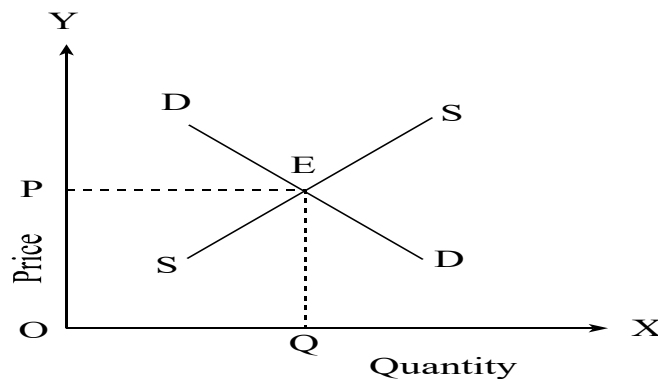
It can be defined as *"Other things remaining the same, as the price of a commodity rises its supply is extended and as price falls its supply is contracted."*

Price of tomato in Rs/kg	Quantity of supply in kg
5	4
10	8
15	12
20	16
25	20



Relationship Between Demand and Supply

Price mechanism is the result of demand and supply force in market. We explain the economic principle of price mechanism by following figure.



Example: 2.1

Suppose the market demand and supply equations for cement are given by

$$Q_d = 40,000 - 20P$$

$$Q_s = 20,000 + 30P$$

- Determine the equilibrium price and quantity (pkt) of cement.
- If increase in housing results a new demand equation of cement; $Q_d = 50,000 - 20P$; What will be the new equilibrium price and quantity of cement?

Solution:

- For equilibrium; there must be; $Q_d = Q_s$

$$40,000 - 20P = 20,000 + 30P$$

$$\text{Or, } -50P = -20,000$$

By solving, we get, $P = 400$ i.e. **equilibrium price of cement will be Rs.400**

$$\text{At } P = 400; Q_d = 40,000 - 20(400) = 40,000 - 8,000 = 32,000$$

$$\text{At } P = 400; Q_s = 20,000 + 30(400) = 20,000 + 12,000 = 32,000$$

Hence, **equilibrium quantity of cement for demand and supply is 32,000 pkt.**

ii) When increase in housing results new demand equation of cement; $Q_d = 50,000 - 20P$

New equilibrium price and quantity would be; $Q_d = Q_s$

$$50,000 - 20P = 20,000 + 30P$$

By solving: $P = \text{Rs. } 600$ and $Q = 38,000 \text{ pkt.}$

2.2 Elasticity, application of elasticity and government policies

Elasticity of Demand

Elasticity of demand is technical term used by the economists to describe the degree of responsiveness of the demand for the commodity to change in price and so on.

$$\text{Elasticity of demand } (E_d) = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in determinant of demand}}$$

Broadly speaking the elasticity of demand is classified into three major types.

- (A) Price elasticity of demand
- (B) Price elasticity of supply
- (C) Income elasticity of demand
- (D) Cross elasticity of demand

(A) Price Elasticity of Demand (E_p)

$$\text{Price elasticity } (E_p) = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in price of the commodity}}$$

$$\text{Symbolically: } E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} \quad \text{OR} \quad E_p = \frac{dQ}{dP} \times \frac{P}{Q}$$

Example: 2.2

Calculate price elasticity of demand when

Price in Rs.	Quantity demand in unit
25	30
15	50

Solution:

$$\text{Here, } \Delta Q = 50 - 30 = 20$$

$$\begin{aligned}\Delta P &= 15 - 25 = -10 \\ Q &= 30 \text{ and } P = 25 \\ E_p &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{20}{-10} \times \frac{25}{30} = -1.66\end{aligned}$$

Here, minus (-ve) sign refers inverse relationship between quantity demanded and price of the commodity. The coefficient $1.66 > 1$; refers elastic demand that means 1 % increase in price results 1.66 % decrease in demand.

(B) Price Elasticity of Supply (PES)

PES = Responsiveness of quantity supply to the change in price.

$$PES = \frac{\text{Proportionate change in supply}}{\text{Proportionate change in price}} = \frac{\Delta Q_s}{\Delta P} \times \frac{P}{Q_s} \quad \text{OR} = \frac{dQ_s}{dP} \times \frac{P}{Q}$$

Example: 2.3

Calculate price elasticity of supply when

Price in Rs.	Quantity Supply
10	1250
8	750

Solution:

$$\begin{aligned}\text{Here, } \Delta Q &= 750 - 1250 = -500 \\ \Delta P &= 8 - 10 = -2 \\ Q &= 1250 \quad \text{and} \quad P = 10 \\ PES &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{-500}{-2} \times \frac{10}{1250} = 2 > 1 \text{ (elastic supply).}\end{aligned}$$

Here, positive relationship between price and quantity supply & coefficient $2 > 1$; refers elastic supply that means 1 % increase in price results 2 % increase in supply.

(C) Income elasticity of demand

$$\text{Income elasticity of demand (E}_Y\text{)} = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in income}}$$

$$\therefore E_Y = \frac{\Delta Q_d}{\Delta Y} \times \frac{Y}{Q_d}$$

$$\text{or, } E_Y = \frac{dQ}{dY} \times \frac{Y}{Q} \quad \left(\frac{dQ}{dY} \text{ is the differentiating } Q \text{ w.r.t. } Y \right)$$

Example: 2.3

Calculate income elasticity of demand when

Income in Rs./Month	Quantity demanded in units
5000	50

6000	55
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Solution:

Here, $\Delta Q_d = 55 - 50 = 5$ and $\Delta Y = 6000 - 5000 = 1000$

$Q = 50$ and $Y = 5000$

$$E_Y = \frac{\Delta Q_d}{\Delta Y} \times \frac{Y}{Q_d} = \frac{5}{1000} \times \frac{5000}{50} \quad \text{Hence, } E_Y = 0.5$$

Here, the coefficient $E_Y = 0.5$ indicates 1 % increase in income results 0.5 % increase in demand. In another words, income elasticity is less than 1 (i.e. less than unitary income elasticity of demand).

(D) Cross elasticity of demand

Cross elasticity of demand; (E_C) = $\frac{\text{Proportionate change in quantity demanded of X}}{\text{Proportionate change in price of Y}}$

$$\text{Cross elasticity } (E_{xy}) = \frac{\frac{\Delta Q_x}{Q_x}}{\frac{\Delta P_y}{P_y}} = \frac{\Delta Q_x}{Q_x} \times \frac{P_y}{\Delta P_y} = \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

Or, $\frac{dQ_x}{dP_y} \times \frac{P_y}{Q_x}$ ($\frac{dQ_x}{dP_y}$ is the differentiating w.r.t. P_y)

Example: 2.4

Compute cross elasticity of demand when

Price of coffee in Rs./cup	Demand for tea in cup
10	40
15	60

Solution:

Let, tea and coffee (substitute goods) are X and Y goods respectively.

Here, $\Delta Q_X = 60 - 40 = 20$

$\Delta P_Y = 15 - 10 = 5$

$Q_X = 40$

$P_Y = 10$

$$E_C = \frac{\Delta Q_X}{\Delta P_Y} \times \frac{P_Y}{Q_X} = \frac{20}{5} \times \frac{10}{40} = 1 > 0 \text{ (positive)}$$

Here, the coefficient $E_C = 1$ shows cross elasticity is positive in case of substitute goods.

Note: The cross elasticity will be negative in case of complementary goods.

Application of Elasticity and Government Policy

Application of Price Elasticity of Demand

- i- Price Determination (product pricing)
- ii- Factor Pricing
- iii- Pricing of Public Utility and Tax Imposition
- iv- Use of Machines
- v- Joint Supply Pricing
- vi- Super Markets Pricing

vii- International Trade (import - export taxed and devaluation)

2.3 Externality and Market Inefficiency

Externality

Externality is a cost or benefit incurred to third party who is not directly involved in an economic transaction and leading to market inefficiencies.

What is positive externality?

Positive externality exists when third party gets benefit without directly involve in market transaction.

For example, **education system** directly benefits to the individuals/students but it also provides benefits to society as a whole through the intelligent behavior.

What is negative externality?

A negative externality exists when third party gets loss without directly involved in the market transaction.

Pollution is a well-known negative externality because emissions of factories directly harm to the natural environment and public health as well as affecting to the nearby communities.

Market Inefficiency

When a market is inefficient: Society is not getting the maximum possible benefit from production and consumption; prices do not reflect true costs or benefits; there is either overproduction or underproduction of certain goods also.

- i- Asymmetry Information:
- ii- Imperfect Competition (monopoly, oligopoly, and monopolistic competition):
- iii- Public Goods:
- iv- Externalities:
- v- Lack of Good Governance:

2.4 Market Failure and Firm Behavior

Market Failure

In economics, *market failure* is a situation in which the allocation of goods and services is not efficient. Hence, it takes place when the market fails to produce desired results i.e. maximizing social welfare.

There are four main reasons why market fails;

i- Market power:

ii- Incomplete information:

iii- Externalities:

iv- Public goods:

Market Inefficiency	Market Failure
Market inefficiency occurs when the market does not allocate resources in the most efficient way, but the market still functions. • Production or consumption is not socially optimal, but the market is still working. • Prices may not reflect true costs or benefits.	Market failure occurs when the market completely fails to produce efficient outcomes, meaning the market system breaks down and cannot allocate resources properly on its own. • Government intervention becomes necessary. • Market forces alone cannot solve the problem.

Firms' Behavior in Market Failure

Market Failure Type	Firms Behavior
Public goods	Do not produce, free-rider problem
Negative externality	Over-produce, ignore social cost
Positive externality	Under-produce
Monopoly	Reduce output and quality, increase price ($P > MC$)
Information asymmetry	Hide information, mislead consumers

UNIT-3 Cost

(2 hours)

3.1 Cost classification

3.1.1 Total, average, fixed, variable, and marginal costs

3.1.2 Direct, indirect, and standard costs

3.1.3 Cash versus book cost, manufacturing and non-manufacturing cost

3.1.4 Sunk cost, opportunity cost, element of cost, life-cycle cost

3.2 Cost estimation and control

3.1 Cost and Its Classification

Cost

- ⌚ The amount of money paid for a thing is called cost or cost is the amount of expenditure incurred on a given thing.

Total cost (TC)

- ⌚ Total money expense for anything / produce.

Average cost (AC)

- ⌚ It is per unit cost for the given level of output.

- ⌚ $AC = \frac{TC}{Q}$

Fixed cost

- ⌚ Those costs which are associated with fixed factors of production like land, plant, building, equipment etc.

Variable cost

- ⌚ Those cost which are associated with variable factors of production and vary with change in level of output like labour, raw materials, fuel etc.

Marginal cost (Incremental cost)

- ⌚ The change in total cost due to change in one unit of extra output.

$$\text{MC} = \frac{dTC}{dQ} \quad \left(\frac{dTC}{dQ} = \text{Differentiating } TC \text{ w.r.t. } Q \right)$$

Direct cost (Prime cost)

- ⌚ It is those costs which can be reasonably measured and allocated with specific output or work activity.

Indirect cost (Overhead cost)

- ⌚ It consists all costs other than direct material, direct labour and direct other expenses.

Manufacturing & Non-Manufacturing Cost

In engineering economics, the term cost is used in various ways because cost is classified according to the needs of management. To understand the cost in a manufacturing company we must have better knowledge about manufacturing and non-manufacturing cost.

Manufacturing cost (factory or production cost)

In the process of converting raw materials into finished goods, a manufacturer needs various natures of costs that refer manufacturing cost. Most manufacturing companies divide manufacturing costs into prime and overhead followed by each cost elements.

a- Prime cost (Direct cost)

Prime costs are those cost related to a given cost object and that can be traced into specific unit of product. Prime cost is the aggregate of the following cost elements.

- Direct material costs
- Direct labour costs
- Direct expenses

Sunk cost

- ⌚ All the past (previous) costs which cannot be recovered when a firm exit from an industry.

Opportunity cost

- ⌚ It is the best rejected opportunity to earn return by using available resources. In another words, it is the best opportunity to generate income that is forgone by not using the limited resources.

Element of cost

On the basis of elements used in production activity, cost can be classified into following three groups: (i) Material cost (ii) Labor cost (iii) Overhead expenses

(i) Material cost

The substance which is used for the production of any things is known as material. Hence, the cost of raw materials, spare parts, consumable materials (fuel/electric, water), packing materials (carton, wrapping paper, cardboard boxes) etc. are called material cost.

(ii) Labour cost

Human effort is indispensable for the conversion of materials into final goods and such effort is known as labour. Hence, all the costs of skilled and unskilled labour employed in construction or production process are called labour cost.

(iii) Other expenses

All other expenses which have not included in material and labour cost is termed other expenses. It includes the cost of special design, drawings or layouts, cost of purchase or hiring of small tools and equipment for a particular job and maintenance costs. Similarly, royalty fee, training expenses, insurance, depreciation, tax, stationary etc. are fall under this category.

Hence, Total cost = Material cost + Labour cost + Other expenses

Life – Cycle of Cost

Life cycle of cost consists following three components:

- (i) First cost (ii) Operating and Maintenance cost (O & M) (iii) Disposal cost

(i) First Cost (Initial or First or Capital Cost)

- (a) Land cost
- (b) Engineering design and development cost
- (c) Building cost
- (d) Basic machine and installation cost
- (e) Training and special tooling cost
- (f) Cost of supporting equipment/computer and transportation
- (g) Manufacturing and fabrication costs etc.

(ii) Operating and Maintenance cost (O & M)

- (a) Labour cost (wages and salaries)
- (b) Material (raw materials)
- (c) Consumable materials (fuel, electric power, water, packing material)
- (d) Overhead items (tax, insurance premium, inventory charges, administrative and management expenses, stationary, small tools, overtime and meeting expenses etc.)
- (e) Transportation cost etc.

(iii) Disposal cost

- (a) Labour cost for removal of the item.
- (b) Material cost (tools, fuel/electric, packing material etc.)
- (c) Transportation cost etc.

3.2 Cost Estimation and Control

Cost estimation can be expressed with the relationship between total cost and its determinants such as the size of plant, level of output, input prices, technology, managerial efficiency, etc.

It can be expressed as; $C = f(S, O, P, T, E, \dots)$

Where; C = Total cost, S = Size of plant, O = Output level, P = Price of inputs i.e. raw materials
 T = Time or Technology, E = Managerial efficiency

Limitations of Cost Estimation

1. Problems in implicit cost (opportunity cost) accounting
2. Change in production technology
3. Change in factor market scenario/competition
4. Change in government policy/ tax rates/ subsidy
5. Estimation of economics of scale
6. Political and natural disturbances
7. Change in foreign exchange rate
8. Pressure of labor union etc.

Cost Control

Cost control is the process of identifying, eliminating or reducing unnecessary [business expenses](#) in order to increase profits. Cost control starts with the budgeting process and looks at vendor selection and negotiation, leveraging [early payment](#) and volume discounts, using [spend management](#) systems, and improving manufacturing or construction efficiency and product quality.

Example-3.1

Following are the costs for the production of 100 badminton racquets;

Labor rate; Rs. 40/hr

Leather; 50 meter @ Rs. 200/meter

Gut; 300 meter @ Rs. 50/meter

Graphite; 100 kg @ Rs. 200/kg.

Total Labor hours needed; 200 hours

Total annual factory overhead; Rs. 50,00,000

Total annual direct labor hours; 2,50,000 hours

Break down the costs into components of prime costs and overhead costs and find out the manufacturing cost of each racquet.

Solution:

For prime (direct) cost:

Direct labor cost = Rs. 40 × 200 hours = Rs. 8000

Direct leather cost = Rs. 200 × 50 meter = Rs. 10,000

Gut cost = Rs. 50 × 300 meter = Rs. 15,000

Graphite cost = Rs. 200 × 100 kg = Rs. 20,000

Total prime cost = Rs 53,000

For the overhead cost;

$$\begin{aligned} \text{Factory overhead per hour cost} &= \frac{\text{Total factory annual overhead cost}}{\text{Total annual direct labor hours}} \\ &= \frac{\text{Rs } 50000000}{250000 \text{ hours}} \end{aligned}$$

$$\begin{aligned}
 &= \text{Rs. } 20/\text{hr} \\
 \text{Total overhead cost} &= \text{Factory overhead per hour cost} \times \text{Total labor hours needed} \\
 &= \text{Rs. } 20 \times 200 \text{ hours} = \text{Rs. } 4,000 \\
 \text{Total cost} &= \text{Total prime cost} + \text{Total overhead cost} \\
 &= \text{Rs. } 53,000 + \text{Rs. } 4,000 = \text{Rs. } 57,000 \\
 \text{Manufacturing cost of each racquet} &= \frac{\text{Total cost}}{\text{Total production units}} \\
 &= \frac{\text{Rs. } 57,000}{100} \\
 &= \text{Rs } 570/\text{racquet}
 \end{aligned}$$

Example: 3.2

The short-run total cost is; $TC = 200 + 5Q - 0.04Q^2 + 0.001Q^3$ and $Q = 10$.

- Find;
- | | |
|------------------------------|----------------------------------|
| (i) Total fixed cost (TFC) | (ii) Total variable cost (TVC) |
| (iii) Total cost (TC) | (iv) Average variable cost (AVC) |
| (v) Average fixed cost (AFC) | (vi) Marginal cost (MC). |

Solution:

Given cost function is: $TC = 200 + 5Q - 0.04Q^2 + 0.001Q^3$ and $Q = 10$

(i) Total Fixed Cost

The cost which is independent of output is called fixed cost, so when the production is zero ($Q = 0$) then the variable cost will be also zero, and total fixed cost will be:

$$TFC = 200$$

(ii) Total Variable Cost

$$\begin{aligned}
 TVC &= 5Q - 0.04Q^2 + 0.001Q^3 = 5(10) - 0.04(10)^2 + 0.001(10)^3 \\
 &= 50 - 4 + 1 = 47
 \end{aligned}$$

(iii) Total Cost

If, $Q = 10$;

$$TC = 200 + 5(10) - 0.04(10)^2 + 0.001(10)^3$$

$$TC = 200 + 50 - 4 + 1$$

$$TC = 247 \quad (\text{Here, } TC = TFC + TVC \quad \text{Or,} \quad TC = 200 + 47 = 247)$$

(iv) Average Variable Cost

When the total variable cost (TVC) is divided by total quantity output (Q) then outcome will be average variable cost.

$$AVC = \frac{TVC}{Q} \text{ or, } \frac{47}{10} = 4.7$$

(v) Average Fixed Cost

When the total fixed cost (TFC) is divided by total quantity output (Q) then the outcome will be average fixed cost.

$$AFC = \frac{TFC}{Q} = \frac{200}{10} = 20$$

(iv) Marginal Cost

Ratio between change in total and total quantity output is known as marginal cost.

$$MC = \frac{dTC}{dQ} = \frac{d}{dQ} (200 + 5Q - 0.04Q^2 + 0.001Q^3) = 5 - 0.08Q + 0.003Q^2$$

$$\text{Hence, } MC = 5 - 0.08Q + 0.003Q^2$$

$$\text{When } Q = 10; \quad MC = 5 - 0.08(10) + 0.003(10)^2 = 5 - 0.8 + 0.3 = 4.5$$

UNIT-4

4.1 Money: Type and Functions

Meaning and Functions of Money

‘Money’ was derived from Latin word ‘Moneta’ another name of Goddess Juno in Roman history.

According to Kent, “Money is anything that is commonly used and generally accepted as a medium of exchange, or as a standard of value.”

Economists considered money has functions four;

Medium of exchange, measure of value, standard of deferred payment and value store.

(i) Medium of Exchange

(ii) Measure of Value

(iii) Standard of Deferred Payments

(iv) Store of Value

Types of Money

(a) Commodity Money

(b) Metallic Money

(c) Paper Money

(d) Bank Money (credit money)

(e) Electronic Money (e-money)

UNIT-5

5.1 Capital Budgeting

Capital budgeting is used by management to plan expenditures on fixed assets. The purpose of capital budgeting is to make long-term investment decisions about whether particular projects will result in sustainable growth and provide the expected returns.

Capital Budgeting refers to the planning process which is used for decision making of the long term investment. It helps in deciding whether the projects are fruitful for the business and will provide the required returns in the future years.

Followings are the steps involved in capital budgeting:

- 1- Identify long-term goals of the business.
- 2- Identify potential investment for meeting the long-term goals identified in Step 1.
- 3- Estimate and analyze the relevant cash flows of the investment proposal identified in Step 2.
- 4- Determine financial feasibility of the investment identified in Step 3 by using the capital budgeting methods like; NPV, IRR, Payback etc.
- 5- Choose the projects to implement from among the investment proposals outlined in Step 4.
- 6- Implement the projects chosen in Step 5.
- 7- Monitor the projects implemented in Step 6 as to how they meet the capital budgeting projections and make adjustments where needed.

Process of Capital Budgeting

1- Identifying and generating projects

2- Evaluating the project

3- Selecting a Project

4- Implementation

5- Performance Review

5.4 Weighted Average Cost of Capital (WACC)

Cost of capital is an important concept in financial management. A firm's cost of capital serves as the linkage between two major financial decisions viz. investment decision and financing decision. **Cost of capital is the required rate of return that must be earned by the company in order to satisfy all their investors.** It is the compensation for time and risk bearing on the project. If a firm earns higher return rate than the cost of capital, then the excess return will lead to an increase in the value of firm's common stock. Similarly, if firm earns a rate of return lower than the cost of capital, the value of common stock decreases.

WACC is the weighted cost of each components of capital structure.

WACC can be obtain by following formula:

$$\text{WACC} = k_{dt} \times W_d + k_{ps} \times W_{ps} + k_s \times W_s + k_e \times W_e$$

Where, $WACC$ = weighted average cost of capital

k_{dt} = cost of debt after-tax $[k_{dt} = k_d (1 - t)]$

k_d = cost of debt (coupon rate)

t = tax rate (corporate income tax)

W_d = weights of debt

k_{ps} = cost of preferred stock

W_{ps} = weights of preferred stock

k_s = cost of retained earnings/internal equity

W_s = weights of retained earnings/internal equity

k_e = cost of external (new issue) equity

W_e = weight of external (new issue) equity

Example: 5.1

Calculate after tax cost of debt while interest rate = 10% and tax rate = 30%.

Solution:

Given, Interest rate (K_d) = 10%

Tax rate (t) = 30%

Cost of debt after tax (K_{dt}) = $K_d (1 - t) = 0.1 (1 - 0.3) = 0.07 = 7\%$

Example: 5.2

If interest on debt is 12%, dividends to shareholders are 15%. Calculate weighted average cost of capital if debt is 70% and equity is 30%. tax rate is 20%.

Solution;

Given,

Interest on debt (K_d) = 12%

Cost of equity (K_e) = 15%

Weight on debt (W_d) = 70%

Weight on equity (W_e) = 30%

Tax rate (t) = 20%

Weighted average cost of capital (WACC) = ?

We know, $WACC = K_{dt} \times W_d + K_e \times W_e$

We know; Cost of debt after tax (K_{dt}) = $K_d (1 - t) = 0.12 (1 - 0.2) = 0.096 = 9.6\%$

$WACC = K_{dt} \times W_d + K_e \times W_e = 0.096 (0.7) + 0.15 (0.3) = 0.112 = 11.22\%$

UNIT-9 Measurement of Nation Income (5 hours)

9.1 Gross domestic product (Components, real and nominal gross domestic product)

9.2 Unemployment (Measurement, job search, minimum wage law)

9.3 Inflation

9.3.1 Causes, effects and measurement of inflation

9.3.2 Constant and current cash flow

9.3.3 Equivalence calculation under inflation

9.3.4 Inflation controlling measures

9.4 Real and nominal exchange rates, fiscal budget and monetary policy

9.5 Financial statement

9.1 Gross Domestic Product (Components, Real and Nominal Gross Domestic Product)

Gross Domestic Product (GDP)

Gross domestic product (GDP) is a measure of all currently produced final goods and services evaluated at market prices within the domestic territory in a year. GDP is the most comprehensive measure of a nation's economic activity. Policymakers use GDP figures to monitor short-run fluctuations in economic activity as well as long-run growth trends.

Hence, $GDP = C + I + G + Nx$ i.e. $(X - M)$ (Measured by expenditure approach/method)

Where, C = Consumption goods (durable goods like- T. V, Car etc. and nondurable goods like- food, cloth etc.) and services (medical care, education, haircuts etc.) bought by households.

I = Investment expenditure made by businesses.

G = Government purchases i.e. the goods and services bought by government.

Nx = Net export (X - M) is the value of goods and services exported to other countries minus the value of goods and services that foreigners provide us when imported.

Real and Nominal GDP

GDP refers to the economic value of goods and services produced within the nation's boundaries in a particular year. When the GDP is estimated at current prices, it exhibits **Nominal GDP**, whereas **Real GDP** is when the estimation is made at constant prices.

Basis	Nominal GDP	Real GDP
Meaning	Current market value of the total output produced within the country's boundaries in a year.	The output produced in a country given period adjusted to the changes in the general price level.
What is it?	GDP with the effect of inflation	Inflation adjusted GDP
Expressed by	GDP = Current output x Current price	GDP = Current output x Base year price i.e. constant price OR: $Real\ GDP = \frac{Nominal\ GDP}{GDP\ Deflator} \times 100\%$
Value	Higher	Generally lower
Uses	Comparison of various quarters of the given year can be made	Comparison of two or more financial year can be done easily
Economic Growth	Cannot be analyzed easily	Good indicator of economic growth

9.2 Unemployment (Measurement, Job search, Minimum wage law)

Unemployment is one of the key indicator of economic health for every country that occurs when individuals who are actively seeking work but unable to find employment. It becomes a serious social and economic issue over the world specially in the under-developed countries like Nepal.

Unemployment rate in Nepal has increased to 10.70 % in 2024 from 10.60 % in 2023. Unemployment in Nepal averaged 10.71 % from 1991 until 2024, reaching high 13.00 % in 2020 due to COVID-19 disruptions and a record low of 10.40 % in 1996 (*Source: National Statistics Office of Nepal*).

Nepal faces unique unemployment challenges due to its economic structure, migration patterns, and limited industrialization. Nepal's economy is dominated by **agriculture** and **informal services**, which cannot provide enough high-quality jobs.

Effects of Unemployment

- Increase in foreign migration for seeking jobs

- **Loss in production and foreign trade**
- **Increase in poverty and economic inequality**
- Increase in foreign dependency
- **Brain drain** (educated youths leaving the country)
- **Lower economic growth and development**
- **Increase in social evils** like- corruption, smuggling, theft, robbery etc.
- People do not trust to government and government failure due to frustration among youths i.e. Gen – Z movement of Nepal in September 8-9, 2025.

It's calculated by dividing the number of unemployed people by the total number of people in the labor force for the particular year.

$$\text{Unemployment Rate of the Year} = \frac{\text{Total unemployed workers}}{\text{Total workers in country}} \times 100\%$$

Minimum Wage Laws

Setting minimum wages is one of the government's key responsibilities that establish a minimum hourly rate that employers must pay their workers. It ensures fair labor standards and protects workers from exploitation in an imperfectly competitive environment. Minimum wages establish the lowest legal amount that employers can pay their workers, helping to guarantee a basic standard of living. This is particularly important for low-income and unskilled workers who may otherwise be vulnerable to unfair compensation.

In many countries, including Nepal, minimum wage laws are also part of broader labor rights frameworks that contribute to decent working conditions, workplace dignity, and overall human development.

Legal Framework for Determining Wages in Nepal

Wages in Nepal are determined and regulated through a combination of constitutional provisions, labor laws, government policies, and institutional mechanisms.

The Constitution of Nepal supports fair labor practices. Article 34 recognizes the right to employment and right to fair remuneration (उचित पारिश्रमिक), laying the foundation for all labor-related legislation.

Government of Nepal has recently revised the minimum wage under the Labor Act, 2017 (2074). This is the **principal law** governing wages in Nepal consisting

- **Minimum wage fixation** for workers and employees
- Wages must be paid **in cash or through banking channels**
- Equal remuneration for **equal work** (no discrimination)
- Timely payment of wages (monthly, weekly, or daily basis)
- Provision of **overtime payment** (generally 1.5 times normal wage)

The official minimum monthly wage for workers in Nepal has been set at **NPR 19,550** per month (**Basic monthly pay**: NPR 12,170 and **Dearness allowance** (महँगी भत्ता / भत्ता): NPR 7,380).

9.3.4 Inflation Controlling Measures

Controlling inflation is essential for maintaining economic stability, safeguarding purchasing power, fostering sustainable growth and the well-being of citizens. It requires a balanced approach involving both monetary and fiscal policies. Central bank can play a critical role through tools such as interest rate adjustments, inflation targeting liquidity management. Here are some key ways to control/manage inflation effectively:

1. Monetary Policy to Curb Inflation

Contractionary monetary policy is a common method for controlling inflation. This approach reduces the supply of money in the economy by lowering bond prices and raising interest rates. When interest rates rise, borrowing becomes more expensive, leading to reduced spending and demand. This helps slow down inflation. Other monetary interventions include managing the money supply, selling securities in the open market, and raising reserve ratios for banks.

2. Fiscal Policy

Fiscal policies, including government spending, public borrowing, and taxes, are effective tools to combat inflation. Reducing government spending can lower public demand for goods and services, curbing inflationary pressures. Increasing taxes on private income reduces disposable income, limiting consumer spending and aggregate demand. A surplus budgeting policy—where the government spends less than it earns in tax revenue—can help manage high inflation rates.

3. Supply Chain Improvements

4. Promoting Competition

5. Subsidies and Price Controls

6. Exchange Rate Policies

7. Encouraging Savings

8. Currency Measures

Currency Demonetisation: Removing high-denomination currency notes can reduce the circulation of unaccounted money, helping control inflation. Such a step is typically taken when the country has a surplus of black money.

New Currency Issuance: Replacing old currency with new notes can address hyperinflation but should be used cautiously as it affects small depositors.

9.4 Real and Nominal Exchange Rates, Fiscal Budget and Monetary Policy

Nominal and Real Exchange Rates

Nominal exchange rate is the expression of one country's currency in terms of another country's currency without adjusting effect of inflation. For example- 1 USD = NPR 144, that means we need Rs. 144 to buy 1 US dollar. It only shows the money-to-money exchange, not purchasing power.

Real exchange rate measures the relative purchasing power of two currencies, after adjusting for price levels (inflation) in both countries.

$$\text{Real Exchange Rate (RER)} = \text{Nominal Exchange Rate} \times \frac{\text{Foreign Price Level}}{\text{Domestic Price Level}}$$

Example:

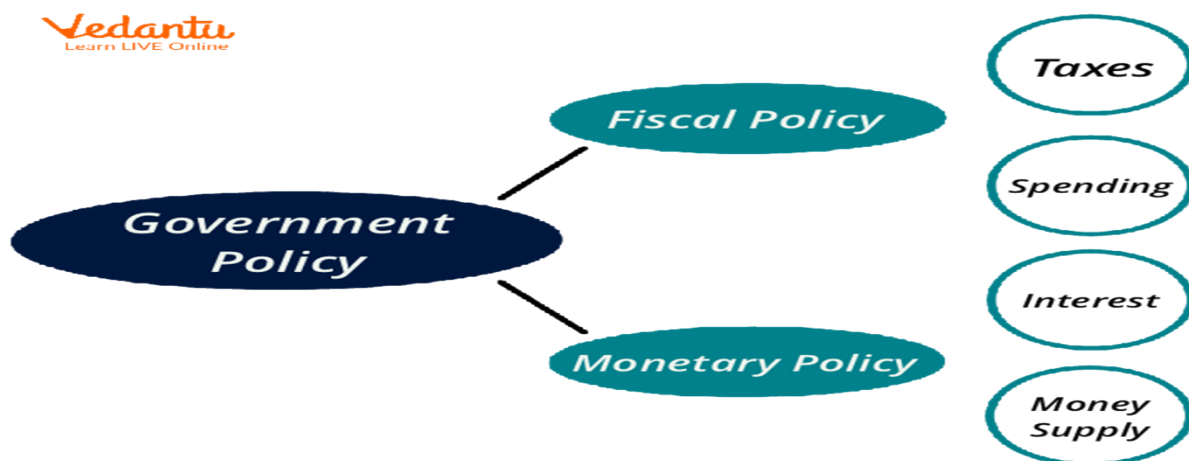
- Nominal exchange rate = 1 USD = NPR 154
- Price level in USA = 120
- Price level in Nepal = 150

$$\text{Real Exchange Rate (RER)} = 154 \times \frac{120}{150} = 123.2$$

Although the nominal exchange rate is 154, after considering inflation the real exchange rate is 123.2.

Interpretation: Goods in Nepal is relatively more expensive compared to the USA.

9.4 Fiscal/Budgetary Policy and Monetary Policy



Fiscal policy of a country is announced by the finance minister through budget every year, so it also termed as budgetary policy. It is a strategy used by government to maintain equilibrium between government receipts through various sources and spending over different projects. There are two types of fiscal policy:

- Expansionary Fiscal Policy: The policy in which the government expands public spending with minimizing taxes (deficit financing).
- Contractionary Fiscal Policy: The policy in which the government reduce public expenditure with increases taxes (surplus budget).

Monetary Policy is carried out by the central Bank on behalf of government to manage the flow of money supply (credit) in the economy. Primary purposes of the monetary policy include bringing price stability, controlling inflation, strengthening the banking system, economic growth etc. There are two types of monetary policies:

- i- Expansionary monetary policy targets high money supplies along with minimization of interest rates.
- ii- Contractionary monetary policy aims to decrease in money supply with rise in interest rates.

The instruments/measures adopted by the central bank to control money/credit in the economy are broadly classified into two categories.

A- General Measures (Quantitative Measures):

- i- Bank Rate
- ii- Cash Reserve Ratio (CRR), Statutory Liquid Ratio (SLR)
- iii- Repo and Reverse Repo Rate
- iv- Open market operations

B- Selective Measures (Qualitative Measures):

- i- Credit Regulation
- ii- Moral persuasion
- iii- Direct Action
- iv- Issue of directives

NEPAL'S BUDGET 2083- 84 (2026-27)

Government of Nepal's Budget for Fiscal Year 2083/84 (2026/27) has emphasized economic reform, good governance, private sector confidence, employment generation, and technology-led development. The main **objectives** and **priorities** are as follows:

Objective	Priorities
i-Achieve high, sustainable, and employment-oriented economic growth.	i- Good Governance and Institutional Reform
ii- Strengthen good governance and improve public service delivery.	ii- Production-Oriented Economy
iii- Restore confidence in the private sector and promote investment.	iii- Private Sector Promotion
iv- Increase production and productivity in agriculture, industry, and services.	iv- Employment Generation
v- Create employment opportunities and reduce youth unemployment.	v- Infrastructure Development
SUMMARY OF BUDGET- 2026/27	
Allocation of Budget Expenditure (in Billions)	Source of Revenue (in Billions)
Current Expenditure = 1270.58 (59.80%)	Tax Revenue = 1405.31 (66.15%)
Capital Expenditure = 431.10 (20.30%)	Foreign Debt = 247.29 (11.64%)
Financial Management = 422.64 (19.90%)	Foreign Grant = 61.74 (02.90%)
Total Budget Allocated = 2124.34 billion	Internal Debt = 410.00(19.30%)
Total = 21 Kharba 24 Arba 34 Crore	Total = 2124.34 (100.00 %)

9.5 Financial Statement

Financial statements contain the basic information about revenue, expenses, assets, liabilities and cash flows during specific period (year). The basic financial statements of any organizations are prepared in accordance with generally accepted accounting principles (GAAP).

Especially income statement and balance sheet are prepared at the end of each accounting period/year. So, they are also termed as year–end statements. But, a cash flow statement may prepare on quarterly, half–yearly or yearly basis as the need of organization.

An income statement comprised of trading account and profit & loss account of an organization that shows profit or loss during a certain period of time (accounting year). Such a profit or loss is ascertained by the comparison of the totals of revenue incomes and expenses of the period.

A balance sheet is termed as a position statement portrays the financial position of a firm up to a given date. Balance sheet shows the details of assets, capital and liabilities and helps to make a comparative study of the relevant items and their changed position between any two consecutive balance sheet dates.

A cash flow statement shows the inflow and outflow of cash of a firm as on a certain date. The position of net cash of a firm as on a certain date helps to plan future activities.

Question	Answered
How much profit did the company make during the fiscal year?	P/L Account i.e. Income Statement
What did the company decide to use their profit for?	Statement of Retained earnings
How much cash did the company generate and spend during the period?	Cash flow Statement
What is the company's financial position up to end of any fiscal period (date)?	Balance Sheet

Objectives of Financial Statement

- i. **Providing financial information**
- ii. **Ascertaining profit/loss**
- iii. **Showing appropriation of profit**
- iv. **Portraying financial position**
- v. **Showing the position of cash**

Importance of Financial Statement

- i. Providing significant financial information to internal and external users like management, trade union, shareholders (owners), debtors, creditors, government and financial analysts for various purposes.
- ii. Ascertaining profit or loss of a firm during a certain period normally a year and facilitating the calculation of profitability in ascertaining performance efficiency.

- iii. Showing the appropriation of profit to different sectors like reserves, dividend etc. and the balance of retained earnings. It is shown by means of profit and loss appropriation account as an important part of the financial statements (i.e. an extended form of profit and loss account).
- iv. Portraying financial position of a firm up to a given date, usually as at the end of each accounting year by means of balance sheet.
- v. Determining the volume of cash and cash equivalent by means of a cash flow statement. It helps to initiate any activity based on the availability of cash. It has now become a common practice of adding a cash flow statement as a part of financial statements, especially in the case of big business entities.

THANK YOU